

KPDS SINAVLARINDA ÇIKMIŞ METİNLER

1. English is now above all an international language, used or understood in most parts of the globe and in a great variety of circumstances. In today's world it serves as a means of expression and communication not only among people who have acquired it as their mother tongue but on an even more remarkable scale with and among non-native users, whose mother-tongues are many and various; and this needs to be recognised in deciding upon the aims and this needs to be recognised in deciding upon the aims and procedures of teaching and learning. English as a world language does not "belong" to mother-tongue speakers of English alone, but to all those who can make effective use of it.
2. The chemicals industry is going through hard times at the moment, and Solvay, Belgium's second largest chemicals firm, is no exception. Last year the company saw profits slide 22 percent, and it is predicting that 1993 profits will be even worse. But strategists at the 150-years-old company's Brussels headquarters are working hard to ensure that the firm is in a strong position to take advantage when economic recovery happens.
3. Since human kidneys work steadily all the time, the best result from a kidney machine naturally comes when it too is in use all the time at a low, steady rate. However, this also has its own drawbacks. One such is that the patient can use machine at the same time. Since it is possible to live for suitable patients to have high-rate dialysis twice a week in place of constant low-rate dialysis. There of course dangers in this practise.
4. Society is, regrettably, less interested in the individual than in the position which he occupies. One almost never asks a person "Who are you?" but one constantly ask "What do you do". For one reason or another, people are assorted into various categories which determine the roles they are to play in society. This assorting process, called social differentiation, goes on in all societies. Women as a class have a status distinct from that of men, and children have a status unlike that of adults.
5. The tuberculosis microbe is airborne which means that anyone sharing a poorly ventilated space with someone with TB, particularly if they are coughing, can contract the illness. Actually, however, only a small fraction of those infected develop active TB since the body's defences normally keep the bug in check. But when the immune system is weakened by age, illness or immune-suppressing drugs, the germ starts to multiply.
6. Those who visit the Mediterranean are invariably impressed with its unity. Everywhere it is the same, for shades of difference here are less important than the resemblance's. Yet this unity is the result of aggressive contrasts-sea and mountain, sea and ocean! In these respects the Mediterranean is very different from either central Europe, or the high tablelands of Asia, Syrian and Syrian deserts, or even the Atlantic Ocean.

7. Mercury has a number of interesting properties and a variety of industrial uses. It expands at a constant rate through the range of temperatures at which it is a liquid. Because of this property and because it does not cling to glass, mercury is often used in thermometers. At ordinary temperatures it evaporates very slowly and can thus be left in an open container for long periods of time. For this reason it is used in one type of barometer. Mercury is a good electrical switches. An electric current passing through mercury vapour causes it to give off light, hence its use in certain kinds of lamps.

8. Protoplasm, which is the fundamental basis of life, is constantly undergoing physical and chemical change. Life, therefore, is the resultant of these constantly occurring changes. There are two great groups into which living things may be classed: plants and animals. Both the plant and the animal kingdoms are very extensive. It is customary, therefore, to regard the science of life under two comprehensive heads, namely, botany which is the study of plants, and zoology which is the study of animals. Both subjects are subdivided into various specialised sections.

9. It is the opinion of most archaeologists that civilisation first developed in the Middle East, where, of all regions of the world, natural conditions offered the greatest assistance to man in his changeover from a life of nomadic wandering as a hunter to settled occupation of the soil. The regular rise of the three larger rivers, Nile, Euphrates, and Tigris; annual renewal of soil fertility by the deposition of a layer of silt: and the generally warm climate favourable both to growth of a rich plant-life, and to the activities of man himself, were all special inducements to the adoption of a way of life based on agriculture.

10. No account of the solar system would be complete without mention of comets, for these are just as much members of the sun's family as are the major and minor planets. Quite a large number of comets are discovered every century, but most of them are extremely faint objects, far bellow the limits of the unaided eye. Comets usually arouse public interest when they are large and bright enough to attract attention and receive mention in the newspapers. But objects of this type do not appear frequently and have been particularly rare during the present century.

11. Many observes, including policy makers, mistakenly assume that economic competition between nations must result in winners and losers. It has been demonstrated, however, that international trade wealth of all participants, even those with lower productivity than their trading partners. The real issue it appears, is the way international competition affects the kinds of goods that any country produces. It seems that the proper role of government is to ensure that its people are prepared to compete in those Industries in which they could or should have an advantage.

12. There are several kinds of dams. One kind is called an earth dam. This consists of a large blank of earth with steep sides. Along the length of the base of the dam a cut-off wall is constructed and this extends down to an impervious stratum. The function of the cut-off wall is to make the foundation watertight. Openings are made in the base of the dam to allow water to escape, thus controlling the level of water. These openings are called sluices. There is one disadvantage with this kind of dam. The material does not become watertight until the reservoir is full of water

13. Today, the United States is in the grip of a sudden industrial revolution. While the first, something from the 1870s to the 1970s, shattered the main section of the American economy from agriculture to industry, the new revolution is shifting the economy away from traditional “smokestack” manufacturing industries to those based upon information, services and new technologies. It took the country decades to accommodate the cultural and social changes resulting from the first industrial revolution, and it would be rashly optimistic to assume that Americans will not face serious stresses in coming to terms with the changes that are transforming the workplace today.

14. A great many books have been written on computers, computer programming, and computer programming languages, particularly fortran. To produce another book on fortran, even the newest fortran IV, probably seems unreasonable to most, and it is with mild trepidation that, I, the author, embark on this project. However, several good reasons can be stated for doing just that. Most computer professionals will agree that the field of computer and information science has quickly become a valid discipline for academia, and that rapid changes are occurring in computer programming languages. Both of these facts demand that a new direction be taken in presenting the subject.

15. Until the late 19th century, most American museums and art academies considered watercolour an amateur pursuit or a preliminary to serious work in oils. Many American watercolorists saw the medium as a holiday diversion, using portable paint boxes and a free style to make what they called “snapshots” of their travels. In contrast, a few recognized the exceptional capacity of watercolour as a medium to provide clear and luminous colours in works that would evoke the ever-changing nature of lakes and rivers they knew so well, and ultimately vie for supremacy with oil paintings in major art collections.

16. Hypothermia, which is more commonly known as “Exposure” by the public, is the state in which the body is unusually low due to having been in severe cold for a long time. Old-aged pensioners can suffer from it in unheated homes with insufficient food. The kind of hypothermia striking climbers affects even much younger people more quickly. When encountered with such a case, urgent action is essential. A hypothermia victim might have been walking quite strongly, but three hours later he could be dead. In treating the victim, it is important to avoid warming the outside of his body, by giving him a hot bath, for example. The aim should be to prevent heat from escaping from the body rather than to try to put heat in back.

17. Human rights” is a fairly new name for what were formerly called “the rights of man”. It was Eleanor Roosevelt in the 1940s who promoted the use of the expression “human rights” when she discovered, through her work in the United Nations, that the rights of men were not understood in some parts of the world to include the rights of women. The “rights of man” at an earlier date had itself replaced the original term “natural rights”, in part, perhaps, because the concept of natural law, with which the concept of natural rights was logically connected, had become a subject of controversy.

18. After 1933 the Western world realized that it was living in another age of absolutism, or rather, in an age of totalitarian dictatorship far worse than the worst of the old absolute kings; such regimes could be seen to be enforcing a “law” that was the command hardly of a “sovereign,” but of a cruel and genocidal despot. It was ordinary people who protested: “This cannot be law. Law, if it is to deserve the name of law, must respect at least some basic rights to which every human being is entitled simply because he is human.”

19. The shopping center emerged in the early 1900s in the suburbs that encircled American cities. Suburbs of that time tended to be chiefly residential and to depend on the traditional city centers for shopping. The first suburban commercial centers had three identifiable features: they consisted of a number of stores built and leased by a single developer; they were usually situated at an important intersection, and they provided plenty of free, off-street parking. These “shopping villages” resembled small-town shopping districts, both in their architecture, which was carefully traditional, and in their layout, which integrated them into the surrounding neighbourhood. The stores faced the street and the parking lots were usually in the rear.

20. Certain features of the motorway undoubtedly ease the strain of driving. Gradients and bends are so controlled as to obviate the necessity of sharp braking and the absence of traffic approaching from the other direction removes one of the commonest sources of accidents. Many dangers remain, however, made more terrible by the high speeds of vehicles. A collision at seventy miles an hour is almost inevitably appalling in its results. A mechanical defect in the car or a puncture can lead to loss of control and catastrophe. The car should be completely roadworthy and tyre pressures and treads need to be checked at regular intervals.

21. The Antarctic is the most remote continent in the world and the last to be discovered, but nevertheless constitutes about one tenth of the world's land surface. So far it has escaped the worst of man's destructive ingenuity but today it is threatened by man's insatiable appetite for natural resources, and seems to be in danger of losing its pristine environment, which serves as the perfect natural laboratory for scientists to pursue knowledge for its own sake.

22. Inflation is a process of steadily rising prices, resulting in a diminishing of the purchasing power of a given nominal sum of money. In other words, you can buy fewer goods for E1 in December than you could in January of the same year. One type of inflation is known as demand-pull inflation. This occurs under conditions of full employment, when demand exceeds supply of goods; that is to say, when people want to buy more goods than are available. The process of demand-pull inflation operates as follows. An increased demand for goods leads to an increased demand for labor, resulting in higher wages and salaries. This has the effect of increasing costs of production and thus causes increased prices. However, as wages and salaries are higher, the increased demand for goods continues and so the cycle goes on.

23. In one century of strenuous research, a vast amount of source material about Michelangelo has been collected, reviewed, edited and annotated, including letters, poems, contracts, receipts, and biographies. Biographical and artistic data have been checked and rechecked, sometimes corroborating and sometimes correcting our previous ideas, and an abundance of new facts have been revealed. Long lost works have been rediscovered and every single known piece has been studied in its formal iconographic, genetic and functional aspects. The artist's character, his daily habits, his working methods, his personal attitudes, and his artistic and political opinions have been traced, as well as the peculiarities of the people with whom he had contact. Thus, modern history of art has formed an image of Michelangelo that is much nearer to truth than those presented by his first biographers.

24. In Eminent Victorians Lytton Strachey portrays four dominating personalities of the nineteenth century. He is, noticeably, free of undue reverence for the great; indeed his satirical view of life enables him to discover in them many flaws which were discreetly overlooked by previous historians. Perhaps his portrayal of General Gordon is the most controversial of all. Certainly, he was a gifted and a gallant soldier, but was he also an unbalanced mystic and a self-opinionated eccentric? His portrait of Dr. Arnold is also disturbing. Was he a wise and for seeing educationalist and headmaster or a tyrant sternly imposing his will on the students in his care? The questions thus raised are intensely provocative and make for stimulating reading.

25. If the key to good nutrition is consuming a variety of foods, then vegetables can truly stand as the cornerstone of a healthy diet. Of all foods, they offer the most diversity. There are literally hundreds of varieties available to us, and because of careful plant breeding, today's vegetable harvest is continually being expanded and improved. In addition, vegetables are replete with nutrients. They supply nearly all of the vitamins and minerals required for good health, and many of them especially starchy vegetables like potatoes and winter squash contain complex carbohydrates, which furnish us with energy. Most also provide dietary fiber, and a few, such as lima beans and potatoes can contribute significantly to our protein intake. At the same time, vegetables contain no cholesterol, have little or no fat and are low in calories in nutritional parlance vegetables are "nutrient dense" - that is their store of nutrients is relatively high for the number of calories they supply.

26. In 1964, the United States Nations Conference on Trade and Development was held. For the first time the poorer nations of the world came together to act as a pressure group on trading matters. The Conference made the following recommendations. The Developing countries should be given free access to world markets for their manufactures and semi manufactures by the elimination of quotas and tariffs. International commodity agreements should be made for each major primary commodity in world trade to stabilize commodity prices. Moreover, compensation schemes, whereby the underdeveloped countries are compensated for the declining prices of their primary products, were recommended for consideration. The conference also resolved that the developed countries should aim to provide at least 1 per cent of their national incomes as aid for the underdeveloped countries.

27. In earlier centuries, it was thought that a great continent must exist in the southern hemisphere, around the South Pole, to balance the known land masses in the north. Its real extent was better understood in the 18th century, particularly when Captain Cook sailed for the first time south of the Antarctic Circle and reached the edge of the icepack. A portion of the ice-covered continent was first sighted by Edward Bransfield in 1820. Explorers of several other nations also sighted portions of the coastline in other quarters and wrote detailed accounts of their observations. However, in the light of these accounts, the first extensive exploration was made by Captain James Clarke Ross in 1841 when a great part of the Antarctic was discovered.

28. Oceanography is the scientific study of the world's oceans, which cover over 70 percent of the earth's surface. The beginnings of modern oceanography go back to the 1870s when, for the first time, wide-ranging scientific observations and studies of the oceans were undertaken by the British. Since then, oceanography has developed into a highly technical and interdisciplinary science, which is now divided into several fields of study. These are biological oceanography, which deals with the study of marine organisms and marine ecology, chemical oceanography, which is concerned with the composition of seawater, and physical oceanography, which studies ocean currents, tides, waves, and other role played by the oceans in climate and weather. Geological oceanography is also another branch of oceanography and is mainly concerned with the formation, composition and evaluation of the ocean basins. Oceanographic knowledge is essential to allow exploitation of the enormous food, mineral and energy resources of the oceans with minimum damage.

29. In 1945, following the Second World War, the allies, that is, the United States, the Soviet Union, and Britain drew up and signed the Postdam Agreement. The main points of this agreement were that militarism and Hitlerism should be destroyed; that industrial power should be so reduced that Germany would never again be in a position to wage aggressive war; that surplus equipment should be destroyed or transferred to replace wrecked plants in allied territories; that Germany should be treated as an economic whole, and that local self-government should be restored on democratic lines as rapidly as was consistent with military security.

30. The police are a regular force established for the preservation of law and order and the prevention and detection of crime. The powers they have vary from country to country and with the type of government: the more civilized and democratic the state is, the less police intervention there is. England, compared with other countries, was slow to develop a police force, and it was not until 1829 that Sir Robert Peel's metropolitan Police Act established a regular force for the metropolis. Later legislation established county and borough forces maintained by local police authorities throughout England and Wales.

31. The International Bank for Reconstruction and Development (IBRD) known as “the International Bank” or as “the World Bank”, is an agency of the United Nations established in 1945. It has the primary function of making loans available to assist developing countries. Usually, loans are made to finance specific projects of investment in underdeveloped countries; and the Bank will normally make a loan only if it is satisfied that the investment will yield a revenue sufficient to enable the payment of interest on the loan, and repayment of the sum lent. In 1983, the Bank made loans to the value of \$3.300 million. Thus a sizable amount of lending is channelled through the Bank, but it is clear that some projects of great value to underdeveloped countries cannot be financed in this way, because they would not yield returns quickly enough or large enough to meet the Bank’s requirements for interest and repayment.

32. The unfavourable effects of cigarette smoking on the heart have frequently been described, but the exact basis for these effects has not been clarified. Some investigators believe nicotine to be the culprit, and there has been some experimental work in animals indicating that large doses of nicotine in conjunction with cholesterol feeding and vitamin D could produce a disease of the arteries resembling that seen in humans. An alternative explanation has been offered by other scientists who have pointed to the possible role of carbon monoxide being inhaled with the cigarette smoke.

33. Agriculture remains the most crucial area for development; here it seems that the most intractable problems of resistance to change exist. One may argue that scientific training in agriculture by itself is unlikely to have any marked impact on agricultural output. Any attempt at vocational training in agriculture presupposes that a meaningful structure of incentive exists for the individual farmer to increase his output, improve his techniques, and expand his range of activities. Without such incentives and opportunities, agricultural education can have little impact.

34. Some decades ago, there was hardly such a subject as the economics of education. Today it is one of the most rapidly growing branches of economics. Together with health economics, it makes up the core of the economics of human resources, a field of inquiry which in the last few years has been silently revolutionizing such traditional subjects as growth economics, labour economics, international trade, and public finance. Consequently, the economics of education with its concept of human investment has rapidly transformed large areas of orthodox economics.

35. Tigers grow to lengths of ten feet or more and can be bigger than the largest lion. They have immense strength. They clutch their prey to them. Holding on with their claws, and depend on the crushing bite of their powerful jaws to end the struggle. They swim very well and can often be seen splashing about in water on very hot days since they apparently suffer from the heat. When the air is chilly, however, they avoid wet or damp vegetation. They can climb, but do not approach the leopard’s ability in this. They can negotiate treacherous rocky areas but generally prefer to stay on level ground. They are not as well equipped with senses as one might expect. They apparently depend on hearing while hunting. Their eyesight is not particularly good, and they seem unable to spot prey until it moves.

36. Scientists have long sought ways to define and measure human intelligence. And while theories of intelligence have grown more sophisticated since the 1800s when some believed mental abilities were determined by the size of a person's head, researchers still do not agree about certain fundamental principles of human thought. They, therefore, continue to debate such basic questions as whether heredity or the environment is more important in forming intelligence.

37. Eliminating poverty is largely a matter of helping children born into poverty to rise out of it. Once families escape from poverty, they do not fall back into it. Middle-class children rarely end up poor. The primary reason poor children do not escape from poverty is that they do not acquire basic mental skills. They cannot read, write, calculate or articulate. Lacking these skills, they cannot get or keep a well-paid job. The best mechanism for breaking this vicious circle is to provide the poor with better educational opportunities. Since children born into poor homes do not acquire the skills they need from their parents, they must be taught these skills in school.

38. Not just in substance but in manner too, Robin Trevelyan, who is the Prime Minister's new right-hand man, is a politician in the old style. He avoids the flourish, which characterizes modern politicians. His speeches are at best unemotional, at worst dull. He is all but incapable of inspiring an audience. His face is inexpressive, solid almost. He evades making promises and is completely lacking in vision. He is a politician whose talent has never been to inspire the mob.

39. Work is central in British culture. When someone asks one "What do you do?", they really mean "What work do you do?" When a woman is asked "Do you work?", what is meant is "Are you doing a paid job?" Yet many people without a paid job work at other kinds of productive activities. Women, notably, perform an unpaid "double shift" in the home as housekeepers and mothers. To confine the term "work" to paid employment, therefore, restricts it far too narrowly. There are many other kinds of work, some of which can take more time and energy than we put into our paid employment from voluntary working in the garden to repairs to the house or the car. In other cultures, work is not as highly valued as this; some people value leisure more, and work only as much as they need in order to provide basic necessities.

40. Alcohol, nicotine and caffeine are psychoactive drugs that are freely available in our society. Their widespread use shows that they provide a common solution to the problems of vast numbers of individuals. The extent and nature of their use is not, however, uniform but varies with the particular sub-culture involved. To take alcohol, for example, there are wide differences between the drinking habits and rituals of merchant seamen and businessmen, between Italians and Jews. Each sub-group in society will have a conception of what is the appropriate situation for a drink; what the permissible and desirable effects of alcohol are; how much it is necessary to drink to achieve this desired state; what is normal and what is deviant drinking behaviour.

41. In the early 1970s, there was a great deal of optimism about improving women's position, ending male privilege and doing away with gender divisions and even gender difference. Equal opportunities legislation was enacted in many countries, and the voice of the women's movement was heard criticizing discrimination between the sexes in every sphere of working life. Now it is clear that legislation can make only a marginal difference to entrenched patterns of job segregation and inequality. The voices of feminism, too, are varied; some demand equality with men, while others pursue the revolution of women's skills and "womanly" virtues.

42. All of us are born, and all of us will die; but there is infinite variety in the nature and circumstances of these two events themselves and in what happens to our bodies and our minds in between. Some individuals, for example, are born without difficulty and grow interruptedly during childhood and adolescence, suffering at worst only minor infectious diseases and accidents. As adults, they reproduce their kind. They age gradually until in extreme old and they die peacefully without pain or discomfort. This is an idealized picture of how we would like things to be, rather than the reality that most people experience. Death comes to many of us, not when we are old, but during or before birth, in infancy, in adolescence, in early adulthood or in middle age.

43. Paper has been known in one form or another from very early times. The papyrus reeds of the Nile swamps served the ancient Egyptians for sheets upon which to inscribe their records. The Chinese and Japanese, centuries later, were using something more akin to modern paper in substance, an Asiatic papermulberry, yielding a smooth fibrous material, being utilised. With the spread of learning in Western Europe the necessity of a readier medium made itself felt, and paper began to be manufactured from pulped rags and other substances. Other papermaking staples were later introduced, such as linen, cotton and wood pulp. The chief raw material in the world paper industry now is wood pulp, the main exporters being the timber-growing countries of Canada, Sweden and Finland.

44. The great expansion in energy demand over recent years has been met to a large extent by petroleum oil. The total world reserves of petroleum oil are still uncertain since large parts of the world are still not fully prospected. The cutback in oil production and the rise in the price of Middle Eastern oil following the 1973 Arab-Israeli war unleashed a worldwide energy crisis, which affected the economies of consumer countries. One result of this crisis has been that Britain has increased its North Sea oil production and become the fifth largest oil producing country in the world.

45. In 1903, the United States signed a treaty with Panama, which gave the United States rights in perpetuity over a 16 km-wide strip of land extending across the narrowest part of Panama for the purpose of building and running a canal. The canal built, now known as the Panama Canal, connects the Atlantic and Pacific Oceans and is just over 80 km. long. Its depth varies from 12 to 26 meters. It is constructed above sea-level, with locks and has been available for commercial shipping since 3 August 1914. An agreement was reached in 1978 for the waterway to be turned over to Panama by the end of the century.

46. When there has been a serious disaster such as an earthquake or flooding, various relief efforts are rapidly put into effect. However, experience has shown that it is usually impractical to attempt mass immunization immediately following a disaster and that, when attempted, it detracts from the overall relief effort without producing a discernible benefit. Effective immunization requires prior planning, good systems of communication and transport, and access to the population at risk. These requirements cannot be met in the immediate post disaster period. Efforts to achieve mass vaccination in the relief phase also drain whatever limited manpower, communication facilities, and transportation exist.

47. Universities are institutions of higher education whose principal objects are the increase of knowledge over a wide field through original thought and research and its extension by the teaching of students. Such societies existed in the ancient world, notably in Greece and India, but the origin of the University as we know it today lies in medieval Europe, the word 'universitas' being a contraction of the Latin term for corporations of teachers and students organised for the promotion of higher learning. The earliest bodies to become recognised under this description were at Bologna and Pans in the first half of the 12th century. Oxford was founded by an early migration of scholars from Paris, and Cambridge began with a further migration from Oxford. Other universities sprang up all over Europe from the 14th century onwards.

48. Romanticism is a term for a movement in the arts, that is, in music, painting, sculpture or literature, which seeks to give expression to the artist's feelings about his subject rather than to be concerned with form or reality. The romantic view is that art is nature seen through a temperament; the realist view, on the other hand, is that art is a slice of life. In painting Delacroix (1798-1863) is the romantic artist par excellence with his uncontrolled expression of the passions and love of the exotic. In literature, the Romantic movement reached its finest form in the works of Goethe, Schiller and Heine; in the poetry of Byron, Keats, Wordsworth, Shelley and Blake; and in the writings of Victor Hugo. Since Romanticism is partly a matter of temperament in the artist just as Classicism is, it may be found at all times and places, although whether or not it becomes predominant depends on contemporary taste.

49. Edison, one of the pioneers of modern technology, lacked formal education. His understanding of literature, art, history and philosophy was superficial. Also, despite the fact that he had invented the phonograph and founded a recording company, his musical taste was abominable. He is therefore sometimes regarded with disdain by academic scientists, who often forget that his ingenuity, inquiring spirit and tireless efforts contributed significantly to the development of modern technology.

50. Under increasing social pressure in the late nineteenth century, some universities opened their doors to a small number of women. More significant, however, was the founding of many women's colleges, frequently run by women. These colleges strove over the years to maintain a curriculum equivalent to that of the largely male universities. Therefore, many leaders of the women's college movement saw themselves as social reformers. Although women entered universities in large numbers in the first half of the twentieth century, their participation was limited by their professional objectives: teaching, social work, nursing, home economics and the like were "women's fields".

51. Atmosphere is the gaseous envelope of the earth, and consists of a mixture of gases and water vapor. The variability of the latter is meteorologically of great importance. The ozone layer, which absorbs solar ultra-violet radiation, especially lethal to plant life, lies between 12 and 50 kilometers above the earth. The lower level of the atmosphere, up to a height of 12 kilometers is known as the troposphere, and it is in this region that nearly all weather phenomena occur. This is the region of most interest to the forecaster studying temperature, humidity, wind-speed and the movement of air masses.

52. For many years after Mt. Everest had been shown to be the highest mountain in the world, political conditions in Nepal, lying south of the summit, and in Tibet, to the north, prevented mountaineers from attempting an ascent. At last, in 1921, the Tibetan authorities gave permission, and the first expedition, organized, as were all subsequent expeditions, by an international joint committee, was sent out. This was primarily a reconnaissance. Besides mapping the northern flank, it found a practicable route up the mountain. By 1936, six further expeditions had climbed on the northern face. Some were hampered by bad weather, others by problems previously little known, such as the effect of high altitudes on the human body and spirit. Nevertheless, notable climbs were accomplished, though the summit was never reached.

53. Edgar Lawrence Doctorow is 55, and on almost anyone's list, he is counted among the leading serious American novelists today. Although he is not simply a writer of comedies, his books sell widely, and three have been made into movies. Readers, some critics excepted, have come to relish the blending of fact and fiction that mark his odd scrutiny of the American past. In his recently published book, *World's Fair*, he turns his historically inventive method on himself drawing heavily on material taken from his 1930s boyhood.

54. William Saroyan's parents and relatives were Armenian immigrants who settled in the farming area around Fresno, California. Saroyan left school at fifteen and went to work doing odd jobs. During this time, he read widely and began writing in his distinctively natural style. By the late 1930s his many short stories, novels and plays had established him as a writer. Many of his stories have been drawn out of his experiences in the Armenian community around Fresno.

55. The Amazon is the largest river in the world. It carries about a quarter of the world's running water and is the second longest after the Nile. Much of it is brown, brackish, piranha-infested and bitterly cold. Ranging from narrow tributaries and raging rapids to stretches of prodigious width and calm, the river's banks can take half a day to reach in parts. It can drop up to 40 metres in less than a kilometre. Furthermore, it runs through deep canyons and step gorges that have been carved out by its turbulent waters.

56. The great window-dresser Gene Moore seems to have been self-taught. As a young man, his main idea was to get away from Birmingham, Alabama, then a town of steel and pollution. It was, he said, the wrong place to be born in for anyone with dreams. He dreamt of being a concert pianist and then of being a painter. But he decided that he did not play very well, and presently gave up painting. In New York in the 1930s, he got various casual jobs. One was with a store that decided he had flair and put him in its display department, and that was the start of his career. He worked for a number of shops promoting their wares and built a reputation for innovative ideas.

57. Pollution is no respecter of national boundaries today. But, environmental scientists can still be surprised by the distances that large quantities of industrial pollutants can sometimes be carried by winds. For instance, a group of chemists at the University of Washington in Seattle have been involved in a case study of such pollutants, which reached the West Coast of America all the way from Asia. They are keen to understand how such an event could take place and to what extent it could have been forecast. In fact, back in March 1997, pollutants such as carbon-monoxide from Asia had been spotted as far across the Pacific Ocean as Hawaii. Thus, it seems increasingly likely that the West Coast of America is particularly exposed to pollution from Asia.

58. Most poetry anthologies are assembled by poets. This is not necessarily a good thing. They are in fact assembled for many different reasons. Some resemble star charts, trying to define the scope of the new and show us what direction poetry is heading. Others turn their gaze on the past, seeking to define poetries of earlier centuries or to identify influential currents of thinking and feeling. Yet other anthologies strive to present enduring images of the beautiful for the reader's pleasure, as if poems were bunches of flowers.

59. Everybody needs vitamins and minerals to remain healthy. The questions are, which ones, how much and when? And the answer is surprisingly simple: take XXX. Actually, the Department of Health has recognised 18 essential vitamins and minerals that we need on a daily basis. The daily amount required of these vitamins and minerals is termed the Recommended Daily Allowance (RDA). XXX meets this requirement and more. As a new vitamin complex, it contains these 18 essential vitamins and minerals, plus a total of no less than 31 other micro-nutrients, including the complete antioxidant group and folic acid. There is no more complete a multimineral-multivitamin on the market. So, because you don't always eat as you should, it makes sense to take XXX.

60. In Japan, there is a government investment and loan programme, known as zaito. Unlike normal government spending, zaito relies not on tax revenues but on people's savings. These are drawn from the publicly-owned postal-savings system, which by law must place all deposits with zaito, and from the postal life-insurance schemes and various pension funds. The finance ministry, which has run zaito for more than 100 years, then lends the money out. During the Second World War, zaito financed Japan's military build-up. Afterwards, it paid for reconstruction and helped to channel low-cost funds into such strategic industries as steel and car-making. More recently, it has turned to "social" investments, such as infrastructure project and housing.

61. Restorative justice does not ask, "How do we punish?", but instead asks, "How do we get people to take responsibility for what they have done?" Paying a fine, or even going to prison are easy options for some people. They are all ways that offenders can avoid taking responsibility because in this way they never have to face the human reality of what they have done. Prisons have been called "universities for criminals". Young people go in for unpaid fines, often for victimless crimes, and they come out with a degree in burglary or worse. I am not saying that the answer is to tear down all prisons. Far from it. There are people who are dangerous to society, whom the community will want to keep locked up. Prison can also be part of a sentencing package under restorative justice. But the vast majority of people in prison are not violent, and do not need to be there. What they do need is to be brought face to face with the human reality of the harm they have caused, and they must be given an opportunity to rectify.

62. In the coming weeks, wine makers north of the equator will oversee the harvesting and fermenting of the first vintage of the millennium. But long before the finished product reaches the shelves – before it even makes it out of the barrel, in some cases – samples will be offered to exporters and distributors. A select group of wine critics will also be given a taste. Most will record their impressions in the extravagant prose that wine journalists unfortunately love to use. Others will go one step further and assign numerical grades. These days a high score is more effective than mere praise, it can make a comparatively unknown wine into a highly desirable one that everyone is seeking to buy.

63. Modern education is almost exclusively focused on preparing children for an urban future, as consumers in a global “free” market. This makes a return to any sort of rural existence almost impossibility for those tutored by the Western education system in the 21st century. The fact is that, for all the fashionable talk about cultural diversity, schools, colleges and universities today prepare their graduates poorly for anything other than a uniform urban existence. We educate the young, from country to city alike, to be urban with urban appetites, skills, minds, dependencies and expectations. And as globalised future will overwhelmingly mean an urban future, our graduates of tomorrow will be trained, above all, to keep the wheels of the global economy running, with all the implications that has for nature and society.

64. People in other European countries have been wondering for some time why and how Norway has stayed out of the European Union. Austria, Finland and Sweden joined in 1994, almost without any public debate, just a few months after their governments had proposed the joining. By then, the Norwegians had been debating the issue for 33 years, ever since their government had started the drive towards unionisation. One reason for the success of Norwegian resistance is that in both 1952 and 1967, when the Norwegian government sent off applications for joining the EEC, President De Gaulle of France rejected the proposals. He feared that the inclusion of Norway, as of England, would complicate and slow down EEC integration.

65. In its full force the Gulf Stream, which begins in the Gulf of Mexico, carries warm water to a depth of up to 100 meters at rates of up to 8 kilometres an hour and penetrates right up into the Arctic Circle to the north of Scandinavia, bearing with it a climate that makes life just about tolerable, even in the thick of the winter. The energy it carries in the form of heat is equivalent to 100 times the entire use of energy in human societies across the world or put another way, more than 27,000 times Britain’s electricity generating capacity. In terms of temperature, the Gulf Stream heats the surface over a wide area by at least 5°C. Were the Gulf Stream to fail, temperatures over northern Europe would fall by more than 10°C during the winter months. Northern Europe would have a climate comparable to that of Siberia: just how it would support its current population is difficult to imagine.

66. Within a short time after the outbreak of the Second World War, Britain was without imports of many vital pharmaceuticals that had formerly come from Japan, Germany and the Far East. As a result, the first wartime government set up systematic research into the cultivation and medical use of herbs. By 1940, women's voluntary organisations had been drawn into a national campaign to gather wild herbs. Up and down the country, County Herb Committees were organised to oversee the gathering, drying, distillation and distribution of the medicinal herbs. Lay people were given brief locally-based training in how to recognise herbs, store and dry them. Farmers were given subsidies to farm certain naturally hard-to-find herbs. By 1943, every county had its herb committee and during the five years of the Second World War, over 750 tons of dried herbs were gathered and turned into medicines.

67. Heat-waves, if the temperature is high enough, above 40°C for instance, lead to wilting, and even death in plant, because of structural damage to essential proteins. The problem is that plants react by closing their pores when, due to a serious heat-wave, they are subjected to water stress, so shutting down on transpiration and conserving water. Just as the body would overheat dangerously if it shut its pores to prevent sweating, so, in a plant, the shutting of the pores will cause permanent damage, if not death. Temperatures above -5°C can damage most plants if lasting for half an hour or more. High soil temperatures will also damage roots and prevent nutrient uptake.

68. A conspicuous feature of cities in many countries, in particular those of Western Europe, is that buildings and streets devastated during the war are, once peace is reinstated, rebuilt in exactly the same manner as they existed before. Enormous efforts are taken to recreate the environment with total fidelity. This reflects the extent to which ordinary people value the traditions and culture of the past. In Japanese cities, however, one sees little evidence of such respect for tradition. Tokyo presents an extreme example: it is quite common these days for the appearance of a street or quarter to change almost beyond recognition every year. In provincial cities as well, one often finds that an absence of several years has rendered a city almost unrecognisable.

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73. Angling is the art of catching fish with very basic equipment, in fact just a rod, a line and a hook, or even just a line and a hook, the special feature of the pursuit being the attraction of the prey by suitable bait. The requisites for a successful angler are knowledge of the haunts and habits of fish, skill in the use of tackle and patience much in excess of that required for most out-of-door sports. Skill in the use of rod and line depends more upon actual experience by the waterside than on acquisition of theories published in books and magazines.

74. The parachute was used for certain kinds of military operation in World War I, but it was not until about 1925 that a conception of airborne forces, that is to say, large numbers of troops moved about by aircraft and deposited at or near the field of battle, by glider, parachute or aeroplane, came into being. Russia was the first to develop the idea on a large scale, and in army manoeuvres in 1930, she conducted practical trials. The Italians were also early in the field with the idea of parachute troops. The French had created an airborne battalion, but it was disbanded before the war. Great Britain had done almost nothing to develop airborne forces up to the outbreak of war in 1939.

75. Letters are often autobiographical records of great importance and some collections of correspondence are practically autobiographies. The preservation and publication of the letters of famous persons is a practice that goes back to antiquity. Thirteen letters ascribed to Plato are still in existence at least some of these are considered by modern scholars to be genuine. Genuine letters of Epicurus and Socrates have also been preserved. Cicero's letters to his friend Atticus, to his daughter and to other correspondents are among the most intimate and interesting autobiographical documents of antiquity.

76. Literature in Australia is a branch of English literature in general, as one might expect from the fact that the population is very largely of British stock and that in education Australia followed English and Scottish traditions as it did in its other institutions. Unlike Canada or South Africa, Australia never had two cultures and two literary traditions, and its isolation during the 19th century meant that its cultural links were almost entirely with Britain. Australians read English books and English magazines for the most part, and 19th – century Australian writers hoped or expected to have their books published in England and wrote mainly with an eye to English readers.

77. An insecticide is a substance employed to destroy insect. It is significant that the word dates from the mid-19th century: only since then has any real progress been made in preventing the attacks and subsequent damage of insects, formerly regarded as unavoidable. Most of the earlier methods were based on the recommendations of such writers as the Roman author Pliny or upon folklore. A few were soundly based on observation and experience, but most were fanciful stories of doubtful logic. The evolution of modern insecticides owes much to gradually improving methods of testing. In early times, natural plant products and minerals were used; later a great variety of industrial by products and synthetic substances became available.

78. Stockholm this year celebrates the centenary of the Nobel Prize, an event that has already been marked by a major exhibition in the city's old Stock Exchange building. The exhibit, "Cultures of Creativity" explores the life and work of Alfred Nobel, inventor of dynamite and one of Stockholm's most illustrious citizens. The prize he established, first awarded in 1901, has subsequently gone to more than 700 scientists, writers and peacemakers for their contributions to humanity. The exhibit also examines the qualities needed to foster creativity and courage, and it explains how nominees and winners are selected.

79. In the case of shallow tunnels or in urban areas it is often possible, by means of carefully sited bore holes to gain an idea as to the nature of the ground and water conditions. Under High Mountain boring becomes expensive so reliance has to be placed upon geological interpretations. As strata can vary so much, surprises are often met with and techniques sometimes have to change in a single tunnel. In the Severn railway tunnel (4 mls 628 yd long, completed in 1886) great quantities of water were unexpectedly encountered and are still being pumped out.

80. Water of doubtful purity for drinking can be rendered safe by boiling and then can be cooled in water bags or in earthenware containers, which must be protected from dust and flies. When boiling is not possible, drinking water can in many areas be adequately sterilized by chlorination; one tablet of Halazone is added to one liter of water and allowed to stand for 30 minutes. Water containing suspended matter should be filtered first. There is, however, the danger of a particularly serious infectious disease in many regions of Africa, the Middle and Far East, and South America. In these regions the water of rivers, lakes and canals may be infected, and the disease is acquired when the water comes in contact with the skin.

81. As with all revolutions, the causes of the American Revolution which separated the original thirteen American colonies from Great Britain were social, economic and political and so inextricably interwoven that it is difficult to appraise them. First there was the distance from Great Britain and the environment of a new country which, whether they willed it or not, had gradually over a period of 150 years turned Englishmen into Americans. The older stock was largely English but the bulk of them, as a contemporary historian commented, "knew little of the mother country, having only heard of her as a distant kingdom, the rulers of which had in the preceding century persecuted and banished their ancestors to the woods of America". With each generation and each move, westward old contacts were broken. Furthermore large groups of colonists had come from Germany, Ireland and other parts of Europe and had no ties with England and, in the case of the Irish, no affection.

82. Translation renders knowledge mobile. The task of the scientific translator, no less than the literary translator, has been to create new texts, to multiply sources into new languages, and thereby to produce new "originals". Over time, translation itself has built a great scientific library, ever more enriched, and accessible. Although we may think of scientific translation as literal, mechanical work, this has never been the case. The reasons for this are complex, but have much to do with the lack of exact one-to-one correspondence among languages. Translating science always involves interpretation, the remaking of an original. If it did not, machine translation would have long ago rendered the scientific translator extinct.

83. The ideal of a family life shared by all in 19th-century England survived into the early 20th century, until home life was seriously dislocated in 1914 by World War I, which was a war on the largest scale the world had ever known. But, since the last decade of the 19th century, new developments and inventions had been rapidly affecting the home life of an increasing number of people. Town and country were knit more closely together by easier railway travel, cheap and efficient postal services, the popularity of the bicycle the development of the petrol engine and the cheap popular newspaper; such things as these helped to break down social formalities and to place women again on a more equal footing with men.

84. Most people take it for granted that prices will always rise, and understandably so. A 60-year-old American has seen them go up by more than 1.000 % in his lifetime. Yet prolonged inflation is a comparatively recent phenomenon. Until about 60 years ago, prices in general were as likely to fall as to rise. On the eve of the First World War, for example, prices in Britain, overall, were almost exactly the same as they had been at the time of the great fire of London in 1666. Now the world may be reverting to that earlier normality. The prices of many things have fallen over the past 12 months or so. Not only computers and video players, but a wide range of goods - from cars and clothes to coffee and petrol - are in many countries, cheaper than they were a year ago.

85. Some people believe that meat consumption contributes to famine and depletes the Earth's natural resources. Indeed, it is often argued that cows and sheep require pasturage that could be better used to grow grain for starving millions in poor countries. Additionally, claims are made that raising livestock requires more water than raising plant foods. But both these arguments are illogical. As for the pasturage argument, this ignores the fact that a large portion of the Earth's dry land is unsuited to cultivation. For instance, desert and mountainous areas are not suitable for cultivation, but are suitable for animal grazing. However, modern commercial farming methods prefer to raise animals in an enclosed space feeding them on grains and soybeans. Unfortunately, the bulk of commercial livestock is not range-fed but stall-fed. Stall-fed animals do not ingest grasses and shrubs (like they should), but are fed an unnatural array of grains and soybeans – which could be eaten by humans. The argument here, then, is not that eating meat depletes the Earth's resources, but that commercial farming methods do. Such methods subject livestock to deplorable living conditions where infections, antibiotics, and synthetic hormones are common. These all lead to unhealthy animals and, by extension, to an unhealthy food product.

86. The chief triumph of this book is its depiction of Wellington. He is not simply the famous British general who defeated Napoleon at Waterloo. He remains a great general but he is also shown to have had feet of clay inside his splendid boots. For example, the writer dwells on Wellington's vanity and his unattractive lack of generosity in sharing the credit for his victories. This is a splendid book. Never less than interesting, but always trenchant. It redefines Wellington without diminishing his achievements and ends by reminding us that it was Napoleon who so forcefully articulated a wish that there should be "a European code of laws, a European judiciary ... one people in Europe". The ogre's dream is coming true.

87. Fast-food is such a pervasive part of American life that it has become synonymous with American Culture. Fast-food was born in America and it has swollen into a \$106-billion industry. America exports fast-food worldwide and its attendant corporate culture, has probably been more influential and done more to destroy local food economies and cultural diversity than any government propaganda program could hope to accomplish. No corner of the earth is safe from its presence and no aspect of life is unaffected. Fast-food is now found in shopping malls, airports, hospitals, gas stations, stadiums, on trains, and increasingly, in schools. There are 23,000 restaurants in one chain alone, and another 2,000 are being opened every year. Their effect has been the same on the millions of people it feeds daily and on the people it employs. Fast-food culture has changed how we work, from its assembly-line kitchens filled with robotic frying machines to the trite phrases spoken to customers by its poorly paid part-time workforce. In the United States, more than 57 percent of the population eat meals away from home on any given day and they spend more money on fast-food than they do on higher education, personal computers, or even on new cars.

88. Even though there have been truly significant advances in modern medicine, health problems still abound and cause untold misery. Although heart disease and cancer were rare at the beginning of the 20th century, today these two diseases strike with increasing frequency, in spite of billions of dollars in research to combat them, and in spite of tremendous advances in diagnostic and surgical techniques. In America, one person in three suffers from allergies, one in ten has ulcers and one in five is mentally ill. Every year, a quarter of a million infants are born with a birth defect and undergo expensive surgery, or are hidden away in institutions. Other degenerative diseases such as arthritis, multiple sclerosis, diabetes, and chronic fatigue afflict a significant majority of Americans. Further learning disabilities make life miserable for seven million young people and their parents. These diseases were extremely rare only a generation or two ago. Today, chronic illness afflicts nearly half of all Americans and causes three out of four deaths in the United States.

89. Does advertising encourage waste by persuading consumers to buy goods that they do not need? In reply to this, it has been pointed out that all the consumer really needs, is a bare minimum of clothing, food and shelter, and that one of the distinguishing marks of any civilized community is that it lives well above the minimum subsistence level. Most advertising is designed to influence the consumer's spending power. In western countries, advertising has played a great part in bringing laboursaving equipment, and so a degree of leisure, and even luxury, to millions. Advertising that encourages the public to want more is also claimed to act as an incentive making people want to earn more in order to buy the goods advertised, and therefore making them work harder. For this reason, advertising has been defended as having an essential part to play in the move towards higher standards of living. The defenders of advertising also point out that it is not solely concerned with encouraging the public to spend. Banks, insurance companies and building societies are amongst the commercial advertisers who encourage saving.

90. Though Italy's national boundaries have altered relatively little since unification in the 1860s, national identity is qualified by sharp internal differentiation. Economic and occupational structures, standards of living, political loyalties, cultural traditions and even language vary substantially between parts of the country. Only since the 1970s has there existed a comprehensive system of regional government with financial and legislative authority. However, the division of powers between central and regional governments is imprecise, and in practice the latter depend on substantial resources from the former. In the absence of clear and effective rules, relations between the regions and the central government are determined by a process of political bargaining. In this process, political alliances and personal linkages play a vital role. In this respect, the Italian system may be defined as a kind of federalism.

91. Sir Philip Sidney was a 16th-century English poet and critic. His *Defence of Poesy* is the only major work of literary criticism in sixteenth-century England, a period during which Italy and France produced large numbers of critical treatises, heavily influenced by Aristotle's *Poetics*. By contrast, Sidney's text is highly eclectic, drawing together aesthetic principles from several traditions and emphasizing especially those principles that are of primary importance to the Elizabethans: ideal imitation, moral teaching and decorum. Looking back to Aristotle, Sidney defines poetry as an imitation of nature, but links that imitation to his view of the poet as maker. The poet imitates not the real nature we see but rather he imitates an ideal nature. Sidney also makes large claims for the didactic role of poetry, following Horace's idea that poetry teaches by delighting.

92. Although the idea of the skyscraper is modern, the inclination to build upward is not. The Great Pyramids, with their broad bases, reached heights unapproached for the next four millennia. But even the great Gothic cathedrals, crafted of bulky stone into an aesthetic of lightness and slenderness are dwarfed by the steel and reinforced concrete structures of the 20th century. It was modern building materials that made the true skyscraper structurally possible, but it was the mechanical device of the elevator that made the skyscraper truly practical. Ironically, it is also the elevator that has had so much to do with limiting the height of most tall buildings to about 70 or 80 stories. Above that, elevator shafts occupy more than 25 percent of the volume of a tall building, and so the economics of renting out space argues against investing in greater height.

93. Land cleared of trees is exposed to erosion, which can be severe in deforested areas having slopes greater than 15 to 17 percent. If land is not disturbed any further and new growth becomes established, erosion may gradually subside. If, however, vegetation on the cutover land is continually removed by man or livestock, erosion will intensify, and environmental problems can be severe. When a forest is removed from a slope, the rate of water runoff is increased two to tenfold or more, depending on the degree of clearing, slope, and rainfall. All too often this leads to flooding of agricultural land in the lowlands. In Pakistan, for example, almost 2 million hectares of standing crops on the lowlands were destroyed by floodwater in 1973, and about 10,000 villages were wiped out. Since valuable soil is lost in floods, the quantity of the arable lands decreases. Alluvial silt deposited elsewhere is rarely usable enough to compensate for such losses.

94. Trade unions, that is, workers' unions, are usually concerned to some extent with mutual benefit activities as well as with collective bargaining and the endeavour to establish standard rates and conditions. The mutual benefit activities have been greatest among the skilled manual workers, whose craft unions have in most cases maintained high rates of contributions and benefits, covering not only dispute benefit but also unemployment, sickness, funeral and often superannuation benefits. The less-skilled workers have not been able to afford the high contributions necessary for such benefits - particularly superannuation - and have usually provided few mutual benefits (except funeral benefit and of course dispute benefit), though some have provided optional benefits in return for higher contributions. In addition to providing cash benefits, most trade unions provide free legal assistance to their members in cases arising out of their employment, and fight important cases affecting their several trades in the courts of law.

95. During the past few decades four East Asian economies – South Korea, Taiwan, Singapore and Hong Kong – have achieved the fastest rates of economic growth the world has ever seen. In 1962, Taiwan stood between Zaire and the Congo on the global ranking of income per head: by 1986 its neighbours were Greece and Malta. In 1962, South Korea was poorer than Sudan: by 1986 it was richer than Argentina. Today the four “dragons” account for 10 per cent of manufactured exports worldwide, not far short of America's 12 per cent. Understanding this miracle is the most urgent task in development economics. But, most economists are content to cite the dragons as proof of their favourite theories – whatever those theories may be. Free marketers point to the dragons' reliance on private enterprise, markets and relatively undistorted trade regimes. Interventionists point with equal assurance to clever bureaucracies, non-market allocation of resources and highly distorted trade regimes.

96. Although women have made huge strides in catching up with men in the workplace, a gender gap still persists both in wages and levels of advancement. Commonly cited explanations for this gap range from charges of sex discrimination to claims that women are more sensitive than men to work versus family conflicts and thus less inclined to make sacrifices for their careers. Now, however, two new studies suggest that another factor may be at work: a deeply ingrained difference in the way men and women react to competition that manifests itself even at an early age. Apparently, females tend to be far less responsive to competition than males – a tendency with important implications for women and business. It may hurt women in highly competitive labour markets, for example, and hamper efficient job placement – especially for positions in which competitiveness is not a useful trait.

97. The Sahara desert takes up most of Egypt's land, so overcrowding is a huge problem. Sixty two million people live squeezed together into the six million fertile acres along the Nile delta and narrow river valley – just five per cent of the total area of Egypt. Between 12 and 15 million live in Cairo alone. Until recently, it was impractical and dangerous to even consider moving into the southern desert, where temperatures regularly rise above 50 °C and water is scarce and can only be reached using carefully placed irrigation wells. But, in the last 20 years a “New Valley” has slowly been taking shape. Towns with industrial centres, tourist areas and spacious apartment blocks are being constructed, factories are springing up. The main development making this possible is the construction of the vast Sheikh Zayed canal, also known as the Toshka canal. Named for Sheikh Zayed al Nahya, president of the United Arab Emirates, which is financially backing the Project, the canal is part of the irrigation scheme dreamed up by the Egyptian government to make it possible for people to move away from the traffic, pollution and bustle of Cairo. If a “second Nile” cuts through the desert and water is distributed to surrounding land, people and crops can thrive there as they do around the existing Nile. The area is becoming known as the New Valley.

98. When Lyndon Johnson assumed the presidency, after the assassination of John F. Kennedy, in November of 1963, he knew that in order to accrue political capital he would initially need to champion goals and policies that Kennedy had already been pursuing. Not long before his death Kennedy had scrawled the word “poverty” on a piece of paper and circled it multiple times; this note fell into the hands of his brother Robert and became a symbolic justification for Johnson's declaration of the war on Poverty, early in 1964. Similarly, many of the things that Johnson pushed through Congress in his first two years as President, can readily be seen as extensions of the avowed policies of the Kennedy Administration. The details might have been different, but historians generally agree that if Kennedy had lived out his first term and won a second, America would have witnessed something similar to the early years of Johnson's Great Society. On foreign policy, too, Johnson at first strove consciously to follow his predecessor. And some historians have argued that in this realm as well, Johnson indeed pursued a course the Kennedy had already introduced. If Kennedy had lived, according to this line of thinking, he would have continued a policy of antagonism towards Cuba and steady escalation of US involvement in Vietnam. Johnson certainly believed that this was what Kennedy intended to do.

99. The space shuttle and its rockets are huge – some 4, 5 million pounds at lift-off. About 85 per cent of that weight is fuel. Since it is designed to work in a vacuum, the shuttle must carry not only fuel but the oxygen to burn it. Because this is an inefficient way to go, NASA engineers have recently tested an engine that gets some of its oxygen on the run. This should reduce takeoff weights by half. A spacecraft equipped with this engine would take off like a rocket. But within minutes, incoming air would begin to supplement liquid oxygen. Once the spacecraft reaches a speed of 1,500 miles per hour – twice the speed of sound- the liquid oxygen would shut off completely and the engine would burn fuel mixed with air. Consequently, the craft would accelerate to about ten times the speed of sound. When the air got too thin for the engine to breathe, the ship would shift back to rocket mode to punch its way into space.

100. It may be that golf originated in Holland but certainly Scotland fostered the game and is famous for it. In fact, in 1457 the Scottish Parliament, disturbed because football and golf had lured young Scots from the more soldierly exercise of archery, passed an ordinance that banned football and golf. James I and Charles I of the royal line of Stuarts were golf enthusiasts, whereby the game came to be known as "the royal and ancient game of golf". The golf balls used in the early games were leather-covered and stuffed with feathers. Clubs of all kinds were fashioned by hand to suit individual players. The great stop in spreading the game came with the change from the feather ball to the present-day ball introduced in about 1850. In 1860, formal competitions began with the establishment of an annual tournament for the British Open championship. There are records of "golf clubs" in the United States as far back as colonial days. However, it remained a rather sedate and almost aristocratic pastime until a 20-year-old Francis Quimet of Boston defeated two great British professionals, Harry Vardon and Ted Ray, in the United States Open championship at Brookline, Mass., in 1913. This feat put the game and Francis Quimet on the front pages of the newspapers and stirred a wave of enthusiasm for the sport.

101. The economic news from Europe was particularly disappointing in the second half of 2002. Moreover, recent surveys from the region imply little prospect of improvement in the near future. Perhaps the most worrying aspect has been the sharp decline in conditions in Germany - the area's largest and most important economy. Domestic demand in Germany is very weak and, with the global economy also struggling, Germany's manufacturers have not been able to export their way out of trouble as they have done in the past. With the economy in such a weak state, it is no surprise then that European stock markets have followed the US stock markets' downturn over the past 6 months. While individual share prices may be lower and market valuations look attractive, the economy does not. Recovery seems some way off and strong equity performance from Europe's markets seems unlikely in 2003.

102. Scientists who study Earth's moon have two big regrets about the six Apollo missions that landed a dozen astronauts on the lunar surface between 1969 and 1972. The biggest regret, of course, is that the missions ended so abruptly, with so much of the moon still unexplored. But, researchers also lament that the great triumph of Apollo led to a popular misconception: because astronauts have visited the moon, there is no compelling reason to go back. In the 1990s, however, two probes that orbited the moon raised new questions about Earth's airless satellite. One stunning discovery was strong evidence of water ice in the perpetually shadowed areas near the moon's poles. Because scientists believe that comets deposited water and organic compounds on both Earth and its moon, well-preserved ice at the lunar poles could yield clues to the origins of life.

103. The US National Institute of Standards and Technology (NIST) will soon be testing a controversial theory about the collapse of the World Trade Center towers. According to an analysis by a leading fire-safety expert, had the fire-proofing insulation on the towers' steel structures been thicker, the towers would have survived longer and might even have remained standing after they were hit by the hijacked planes. The work is being seized on by lawyers representing victims' families and insurance companies. If confirmed, it could also lead to changes in building codes. NIST is responsible for drawing up the final report on the towers' collapses and recommending if any changes are needed. It is widely accepted that the collapses were caused by the failure of the buildings' steel structure as it was weakened by the heat of the fires.

104. The long-expected decline in the dollar is now well under way. For years, economists have predicted that America's huge current-account deficit would eventually cause its currency to plunge. So far, the dollar's slide has been fairly gradual: it is down by 13% in trade-weighted terms over the past year, though it has dropped by almost twice as much against the euro since its 2001 peak. As the decline seemed to pick up speed this week, John Snow, George Bush's Treasury Secretary, declared that he favours a "strong dollar policy". That was surely the wrong answer, even leaving aside the debatable issue of whether cabinet secretaries can influence the level of exchange rates. A weaker, not a stronger dollar, is what the world needs now - so long as policymakers elsewhere respond appropriately. America promoted a strong dollar throughout the 1990s, when inflation was still thought to be the main enemy. Today it makes less sense. Even after its recent slide, the dollar seems overvalued. Moreover, with ample spare capacity in America, deflation looks a bigger risk than inflation.

105. Why are people prejudiced? Not surprisingly, theories of prejudice have tended to focus on the more extreme forms of prejudice, in particular when there is aggression and violence. At the turn of the last century, it was popular to consider prejudice to be an innate and instinctive reaction to certain categories of person (e.g. certain races) much as animals would react in instinctive ways to one another. This sort of approach is no longer popular, as it doesn't stand up well to scientific scrutiny. However, there may be an innate component to prejudice. There is some evidence that higher animals, including humans, have an inherent fear of the unfamiliar and unusual, which might set the mould for negative attitudes towards groups that are considered different in certain ways. There is also evidence for a mere exposure effect, in which, people's attitudes towards various stimuli (e.g. other people), improve as a direct function of repeated exposure. Exposure to familiarity with the stimulus, provided, that initial reactions to the stimuli are not negative. Another perspective rests on the belief that prejudices are learned. Indeed, it has been argued that hate and suspicion of certain groups are learned early in life, before the child even knows anything about the target group and that this provides an emotional framework that colours all subsequent information about, and experience with, the group.

106. By the early 19th century, the eminent French zoologist Georges Cuvier believed he had found rock-solid evidence for the biblical great flood. While studying the geological strata around Paris, Cuvier found that fossils of sea creatures in one ancient layer of chalk were overlaid by those of land creatures. Then just as abruptly the layer above contained sea creatures again, with the top layer showing evidence of a vast and rapid inundation around present-day Paris. Cuvier regarded these sudden changes in the fossil record as evidence for sudden Catastrophes which devastated life on Earth, of which the great flood was just the most recent example. Cuvier's discoveries, published in 1812 won support from a large number of eminent scientists such as the geologist Sir James Hall. However there were a few who were deeply sceptical, pointing out that the evidence of a global flood was far from conclusive. Most sceptical of all were the followers of the Scottish geologist James Hutton. In 1795, he had published a two-volume text based on the view that the slow steady processes that shape our planet today, such as erosion, were also crucially important in the distant past.

107. No child is too young to play and therefore to engage in engineering, even though it is of a primitive kind. We all did so as children ourselves when we devised our own toys and games and sometimes even imaginary friends to enjoy them with us. The idea of playfulness is embedded in engineering through the concepts of invention and design. Not that engineering is trivial; rather, the heart of the activity is to give imagination its freedom to dream and turn those dreams into reality. Children do experience the essence of engineering in their earliest activities, yet there is seldom any recognition that this is the case. They may hear the word "engineer" only in connection with railroad locomotives and have no idea that their playful activity could become a lifelong profession. Engineers themselves are understandably reluctant to equate their professional activity with mere child's play. After all, they studied long and hard to master complicated knowledge of atoms and molecules, stresses and strains, heat and power, current and voltages, bits and bytes. They use computers for serious modelling and calculation, not for fun and games. They design and build real towers and bridges that test the limits of reliability and safety, not toy ones that totter and fall down with little consequence.

108. Europe and Japan do not use fuel economy standards to any significant degree, but instead rely principally on high taxes to reduce gas consumption. Their average tax is more than \$2 per gallon, while in the US, federal gas taxes are only 18 EURO per gallon and average state taxes 22 euro per gallon. Higher prices at the pump resulting from higher taxes increase consumer demand for cars with better fuel economy. They also encourage consumers to reduce their driving. Research shows that federal taxes on gasoline would have to increase by a bit less than 50 euro per gallon to cut gasoline consumption in the US. Although a 50-euro increase is a lot compared with the present average total tax of 40 euro, it would raise retail gas prices to only a little more than \$2 per gallon, tax included. This is far below prices in Europe and Japan. Even if federal taxes on gas were doubled, US retail gas prices would still be much below those in other developed nations.

109. Throughout his working life, Shakespeare worked as an actor in the midst of a troupe. We know little about his first years in London. For a few years between 1585 and 1592 his name disappears altogether from the public records, and the most likely reason for this is that, for at least some of this time, he was working for one of the city's acting companies; as a junior member he would not be listed among the troupe's principal players. In the late 1580s, theatrical activity in London was largely concentrated in Shoreditch and Southwark, districts of London. Shakespeare could have lived anywhere, but Shoreditch, which would have been cheap and convenient, is a likely candidate for a young actor. In his early career, Shakespeare may have moved from troupe to troupe in order to survive. Whatever the case, working conditions must have been similar. Sundays, religious holidays and disasters aside, a company would perform a different play each afternoon of the week, though some plays would be repeated in the weeks ahead. An actor usually had to keep at least 30 parts in his memory and a leading player such as Alleyn or Burbage must have kept in mind nearly 5,00 lines a week.

110. Behavioural biologist Jane Atkinson and her colleagues have been studying the subtleties of how crows steal food from one another. Atkinson had been watching the birds at the beach as they fed on fish, clams and other small animals in the intertidal zone. She noticed that if a crow had found a particularly large meal that could not be eaten in a single gulp, another crow would often come by and try to steal the food away. Food theft is fairly common in the bird world, so the crows' thievery wasn't unexpected. What really intrigued Atkinson was that the birds employed two different tactics to take the food? In some instances, the thieving bird would take an aggressive approach - typically involving some chasing or physical contact, such as pecking in other exchanges, however, the thief would use a more passive method: merely approaching the other bird secretly and stealing the food without any commotion at all. What the team wanted to know was: how did these tactics fit into the group foraging practices of the crows?

111. In many ways, Hollywood seems to exemplify the most joyless aspects of capitalism. The "industry", as it insists upon calling itself, packages artistic ideas and images as commodities and then values those commodities according to how they "penetrate" markets. The system's worrying inefficiency, of course, is that studios never know what the public at large will want to buy. So, films are tested in front of preview audiences, revised according to the audience's suggestions, tested again, and then marketed with a vigour directly proportionate to the test scores. There are two problems with this approach. The first is that the test-sample size is minimal but can determine a film's fate. The second is that by the time the test audience sees a film it's too late to change it very much anyway, particularly when twenty, fifty or a hundred million dollars has already been spent.

112. Reading presents a real paradox to neurobiologists. It was only invented a few thousand years ago, so there really has not been enough time for our brain to evolve specialized ways to do it. How do brain circuits produced by millions of years of evolution in a world without written words adapt to the specific challenges of reading? We know we have to learn the skill but how does our brain learn to read? In the social sciences, the majority of researchers do not see a problem. There is a widespread view that the brain is a completely adaptable organ, capable of absorbing any form of culture. Yet recent findings from brain imaging studies and neurophysiology throw new light on the organization of the reading circuits in the brain. The findings contradict this simplistic model of a brain that merely absorbs everything from its cultural environment. And, they suggest that the architecture of our brain is limited by strong genetic constraints though it seems that it has still some degree of flexibility.

113. Throughout history, eyewitnesses have reported orange glows, fireballs or flashes in the days before and during an earthquake. It was in 1968, however, that the first photographs of "earthquake lights" were taken during a series of earthquakes in Japan. Some showed red streaks across the sky. Others looked like a low blue dawn from a distance. In 1999, floating balls of light in the sky were broadcasted on Turkish television, reportedly filmed the night before the devastating earthquake of 7.4 on the Richter scale that killed many thousand people in the Marmara region of Turkey. Mysterious or not, repeated sightings of earthquake lights confirm their existence. It has to be said that earthquake lights are a fairly well-known phenomenon, but we don't know what they mean, or what causes them. Seismologists have struggled far years to find a reliable earthquake predictor. Could the lights hold the key?

114. Much has been said and written about the declining numbers of and disappointing lack of diversity among American college students majoring in engineering. Among the factors cited to explain this phenomenon are the lack of exposure of high school students to the very idea of engineering and the fact that many have insufficient mathematics and science background to gain entrance to engineering school, even if they do identify the profession as a possible career. This is unfortunate, for the ideas of engineering should be integrated into the curricula not only of high schools but also of middle and primary schools. Our children are being done a disservice by not being exposed properly throughout their education to engineering activities identified as such. After all, even pre-school children have the prerequisites in their play for appreciating exactly what engineering is: design. Indeed, design is everywhere around them throughout their school day, even in their before-school and after-school activities. It need only be pointed out to them that they are designing something, and therefore being engineers of sorts, in virtually everything that they do.

115. For two decades after World War II, mass production reigned supreme. Mass-production techniques pushed companies into standardized products, long product life cycles, and rigid manufacturing, emphasizing efficiency and low cost over flexibility. Special orders cost more. But, today's consumers are very choosy. They want quality, value and products specially tailored to their needs, but always at the lowest possible price. For now, mass customization has come to the fore. Mass customization uses information technology to produce and deliver products and services designed to fit the specifications of individual customers. Companies can customize products in quantities as small as one with the same speed and low cost as mass-production methods. Mass-customization systems use information taken from the customer to control the flow of goods.

116. Before the Polish-born French-American mathematician Benoit Mandelbrot made his mark on the world, scientists liked to forget about the imperfections and irregularities of nature. The study of perfect squares, triangles and planes had dominated their field for over 2,000 years, since the Greek geometer Euclid wrote maths' oldest treatise "Elements" and provided us with the tools to measure these flawlessly smooth shapes. Early question about how to measure the real shape of a tree, a coastline or anything with a rough edge could not be tackled by Euclidean geometry and had therefore been ignored. But, Mandelbrot changed all this when he invented fractal geometry, which enables us to measure roughness. "My whole career has been one long, ardent pursuit of the concept of roughness", he says. "The roughness of clusters in the physics of disorder, of turbulent flows, of exotic noises, of chaotic dynamical systems, of the distribution of galaxies, of coastlines, of stock-price charts and of mathematical constructions."

117. Therapists have to be very careful before they make a diagnosis of delusional disorder. A great many complaints are founded on fact. It is possible that a patient is really being harassed at work, that her husband is deceiving her, or that her business partner is cheating her. Indeed, therapists must be careful not to mislabel facts as delusions, a trap known as "the Martha Mitchell effect". Martha Mitchell was the wife of former US attorney general John Mitchell. In October 1972, he was accused of having ordered the break-in at the Democratic campaign headquarters in the Watergate Hotel in Washington, D.C. Mrs Mitchell repeatedly told the press that her husband was being made a scapegoat to protect the real culprit - President Richard M. Nixon. The White House spread disinformation about Mrs Mitchell, saying she had a drinking problem and implying that her statements were delusional. When the scandal was ultimately unravelled, Mrs Mitchell's statements were proved true and she was shown to be utterly sane and with no drinking problem.

118. Recent activity in several US church communities has seemed almost unbelievable: churchgoers have gathered around huge fires and cheered as they cast Harry Potter books into the flames. They fear that the incredibly popular series about a school for young wizards is spurring children and adolescents toward a life of witchcraft and onto the dangerous path toward Satanism. For these congregations, J.K. Rowling's books are none other than the work of the devil herself. To most people, however, the Harry Potter books and films are merely compelling adventure stories, not a threat to children's psyches. But what has been forgotten in the excitement of "Pottermania" is that boys and girls have been fascinated by magic and sorcery for generations. Surveys about magical practices among adolescents vary widely, but some indicate that as many as 44 per cent have shown some slight, passing interest in it. Although satanically motivated violence occasionally makes headlines, research shows that less than 5 per cent of young people take part in more extensive witchcraft, and very few end up in the kind of organized devil worship that can lead to such acts as ritual murder.

119. Why does seawater taste salty? It is a question that has been asked by countless people down the ages. And, the answer seems straightforward: rain constantly erodes the surface of the Earth, washing a mix of natural chemicals into rivers and thence into the sea. The most water-soluble and abundant of these just happen to taste salty. All very simple. Or is it? After all, erosion has been taking place for millions of years, dumping ever more of these salty compounds into the sea, yet the concentration is still far below the saturation level. So, the real mystery is not why the sea tastes salty, but why it isn't utterly packed with salt, and as lifeless as the Dead Sea. Here is another curious thing about our planet. Its atmosphere has existed for billions of years, and yet it still contains a mix of highly reactive gases like oxygen and methane. Why haven't they settled down into a boring unreactive atmosphere like that of Mars or Venus?

120. Family-owned companies are bad for business, a new study argues—at least when they dominate a large chunk of a country's economy. Outside the United States and the United Kingdom, most major corporations are in the hands of a few wealthy families, rather than being owned by a wide network of shareholders. The power of these small families often extends far beyond the companies they own directly, thanks to a system of "control pyramids" in which they exercise indirect control over a slew of smaller companies. This concentration of corporate power doesn't merely leave a high percentage of wealth in the hands of billionaires—it also retards growth, diminishes efficiency, and limits economic freedom. Moreover, "a tiny elite that cannot be sacked," as the study puts it, is likely to pursue "economic entrenchment," in which property rights and financial openness are curbed to protect a few families' economic and political prerogatives.